
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* research has tried to improve reasoning by inserting trained vectors into LLM inputs, yet whether the gain comes from the *learned content* or from the *act of injection itself* has not been carefully separated. We study Random Soft Prompts (RSPs), which drop the training step entirely and append a freshly drawn sequence of random embedding vectors to the input. Each RSP vector is sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table; the sequence carries no learned content, and yet matches the accuracy of trained soft prompts on math reasoning benchmarks. The mechanism unfolds in two stages: because attention has to absorb a never-seen-before random position, the distribution over the first few generated tokens flattens and reasoning trajectories *branch*, and as generation continues this influence dilutes naturally so the response *converges toward the goal*. We show that during inference RSPs lift early-stage token diversity and, combined with temperature sampling, widen Pass@ N , the probability that at least one out of N attempts is correct. Beyond inference, we carry the same effect into DAPO training and demonstrate practical gains. Our contributions are: (i) a new finding that purely training-free random embedding injection alone drives latent reasoning; (ii) a theoretical and empirical validation of the underlying mechanism; and (iii) an extension from inference to training.

1 Introduction

As large language models (LLMs) scale, full fine-tuning becomes prohibitively expensive, motivating parameter-efficient methods that adapt frozen models with few extra parameters [Hu et al., 2022, Houshy et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input, steering behavior through attention [Li and Liang, 2021, Lester et al., 2021]. The paradigm is actively adopted for LLM mathematical reasoning [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]: despite diverse designs, these methods share a common structure of injecting out-of-vocabulary continuous embeddings and *optimizing* them for downstream performance. Such optimization is task- and dataset-specific, and prior work generally interprets the resulting gains as a consequence of the trained values of these embeddings.

However, we observed that optimization is not necessary for these effects: injecting untrained random embeddings can also lift downstream performance. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning, but appending random embeddings to such mismatched inputs mitigates this degradation. If simple noise were merely shaking the model, accuracy should worsen instead. Therefore, the injection itself, independent of any learned content, alters model behavior in non-trivial ways. Existing studies focus on end-task accuracy and have not separated the effects of injection from those of optimization, leaving how embedding injection reshapes the output distribution insufficiently explored.

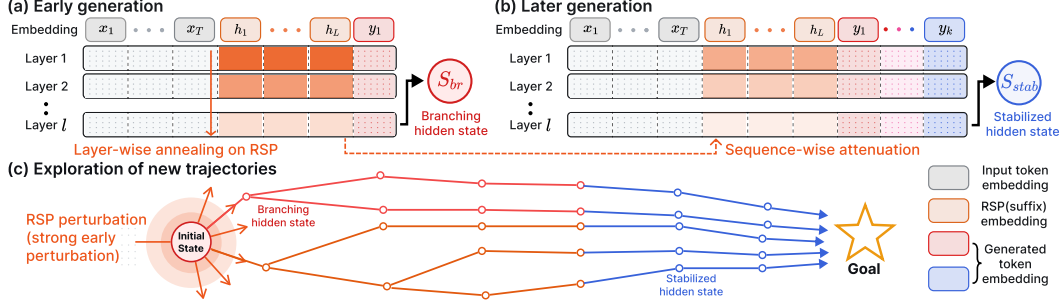


Figure 1: Conceptual overview of RSP-induced trajectory diversity. The hidden states shown are the final-layer outputs that drive the next-token distribution. (a) Early decoding: RSP perturbation reaches a *branching state* (red; formalized in §3.1), whose top- K differs from the no-RSP baseline and induces a diverse output distribution. (b) Later decoding: as the KV cache grows, the bound on RSP attention mass narrows (Theorem 1); branching events become rare and the hidden state (blue) commits to the branch already opened. (c) Across rollouts, the trajectories opened by early RSP perturbation accumulate into independent paths that converge to goal-directed completions.

We analyze the effect of training-free Random Soft Prompts (RSPs) on LLM reasoning. RSPs are continuous vectors drawn from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table, with each rollout receiving an *independent* draw that enables broader exploration. The benefit appears most strongly during early decoding where wider exploration is needed: a single RSP injection induces hidden states that *branch* into diverse trajectories, and as decoding deepens these trajectories *stabilize* into goal-directed completions. Empirically, RSPs lift early-stage entropy and, combined with temperature sampling, expand the Pass@ N trajectory diversity, and we further show that the same exploration benefit carries from inference into DAPO [Yu et al., 2025] training. Our contributions are: (i) the new finding that *training-free* random embedding injection alone drives latent reasoning capabilities, suggesting much of the effect attributed to learned soft prompts may come from the *act of injection itself*; (ii) theoretical and empirical validation of the mechanism; and (iii) an extension from inference to training, demonstrating practical applicability.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Existing *soft prompt* methods all share a common structure: they inject out-of-vocabulary continuous embeddings and *optimize* them under task-specific objectives. Because evaluation is anchored to end-task accuracy, the contributions of “injection itself” and “optimization” are not separated, and how plain embedding injection reshapes model behavior remains an open question. As parameter-efficient alternatives to full fine-tuning, Prefix-Tuning [Li and Liang, 2021], Prompt Tuning [Lester et al., 2021], and P-tuning [Liu et al., 2024] reached near-fine-tuning performance using learnable continuous vectors, after which LLM-reasoning-specific variants quickly followed. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via trajectory entropy minimization; SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model; LTPO [Ye et al., 2026] optimizes the soft prompt per problem to maximize its own confidence on that problem; COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input embeddings; MemGen [Zhang et al., 2026] uses embedding vectors as experience memory; and Pause tokens [Goyal et al., 2024] insert learnable tokens to provide additional computation steps.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at several stages and locations. During training, dropout [Srivastava et al., 2014] randomly deactivates *hidden activations*, and NEFTune [Jain et al., 2024] *adds* Gaussian noise to *input embeddings* during instruction fine-tuning to improve instruction following. At inference time, randomized smoothing [Cohen et al., 2019] injects Gaussian noise into *raw inputs* to obtain certified robustness, and SmoothLLM [Robey et al., 2025] perturbs prompts at

71 the *discrete character* level to defend against jailbreaking. In RL, NoisyNet [Fortunato et al., 2018]
 72 adds learnable, trained noise to *network weights* to aid exploration.

73 RSP appends noise to *embeddings*, and differs from these in three respects. (1) It operates without
 74 any training, purely at inference, and does not modify model parameters (unlike dropout, NEFTune,
 75 and NoisyNet). (2) It preserves the original input and appends the embedding within a single forward
 76 pass, so it does not require the “multiple noisy passes plus aggregation” that robustness methods
 77 need. (3) It uses vectors drawn directly from the pretrained embedding statistics, with no learned
 78 parameters at all (NoisyNet, by contrast, learns its noise parameters).

79 3 Random Soft Prompts and Why They Work

80 RSP does not add to the hidden state directly but enters through self-attention, so within one attention
 81 head its instantaneous influence reduces to a single scalar: the *attention mass* w on RSP tokens. This
 82 section tracks how w evolves during decoding. Early on w is large enough to branch the hidden state
 83 into multiple directions (§3.2); as the KV cache grows, its upper bound narrows and the response
 84 converges (§3.3). Randomness supplies *independent* perturbations across rollouts so that each opens
 85 a fresh branch. Further proofs are in Appendix D.

86 3.1 Random Soft Prompt

87 We start with the object being analyzed. Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the pretrained token-embedding
 88 matrix, with V the vocabulary size and d the hidden dimension. Write its entrywise statistics as
 89 $\mu_E := \frac{1}{V \cdot d} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ (the arithmetic mean over all $V \cdot d$ entries) and $\sigma_E := \text{std}(\mathbf{W}_E)$. An RSP
 90 of length L is a sequence of continuous vectors $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ drawn independently
 91 from

$$h_j \sim \mathcal{N}(\mu_E \mathbf{1}_d, \sigma_E^2 \mathbf{I}_d), \quad j = 1, \dots, L. \quad (1)$$

92 The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. Letting
 93 $\mathbf{X} \in \mathbb{R}^{T \times d}$ denote the input token-embedding sequence (the T rows of $\mathbf{W}_E$ corresponding to the
 94 prompt tokens), the main text uses the *suffix* form $[\mathbf{X}; \mathbf{H}] \in \mathbb{R}^{(T+L) \times d}$ as the default; other positions
 95 (prefix, infix [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]) are reported as
 96 ablations in §4.2 and Appendix A. RSP is not a learnable parameter, so it receives no gradient and is
 97 freshly drawn for each rollout.

98 Call decoding step t a *branching event* and the last-layer hidden state $\mathbf{s}^{(t)}$ a *branching state* when
 99 its top- K differs from that of the no-RSP baseline $\bar{\mathbf{s}}^{(t)}$ (Figure 1, red).¹ §3.2 asks *when* a branching
 100 event is reachable, and §3.3 asks *how* that reachability fades.

101 3.2 Exploration: RSP strengthens early-stage exploration

102 Inserting a random prompt concentrates attention on it, and that concentration is what amplifies
 103 exploration. The effect of a single RSP injection on the hidden state can be examined along two
 104 questions: *how much*, and *in which direction*. The first is decided by a single scalar w inside one
 105 attention head, the second by the distribution of $\mathbf{H}$. RSP’s design choice, an isotropic Gaussian,
 106 addresses both at once.

107 Consider one attention head in self-attention layer ℓ . At query position i , the query attends $n \geq 1$
 108 unmasked real tokens together with $L \geq 1$ RSP tokens, all attention logits finite. Write the attention
 109 weights as $\alpha_{ij}^{(\ell)}$ and the value vectors on the real and RSP sides as $\mathbf{v}_j^{(\ell)}, \mathbf{v}_{r,j}^{(\ell)}$. Defining the total
 110 attention mass on random tokens $w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$, the real-token mass $1 - w_{r,i}^{(\ell)}$ is positive
 111 and the head output $\mathbf{o}_i^{(\ell)}$ splits *exactly* into a renormalized real-token term $\tilde{\mathbf{o}}_i^{(\ell)}$ and an RSP-induced

¹Italic h_j lives in the RSP input space; bold $\mathbf{s}^{(t)}$ is the model’s hidden state. K is the next-token sampler’s candidate-set size, instantiated as either the fixed cutoff of top- K truncation or the dynamic nucleus size of top- p sampling. t indexes decoding step, ℓ indexes self-attention layer.

112 contribution $\boldsymbol{\eta}_i^{(\ell)}$:

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}. \quad (2)$$

113 *Derivation of Eq. (2).* Write the head output as $\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}$, then factor
 114 $1 - w_{r,i}^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} > 0$ out of the real-token sum. No approximation beyond the softmax-
 115 normalization identity is used. $\square$

116 This single quantity $w_{r,i}^{(\ell)}$ controls both the attenuation ratio $1 - w_{r,i}^{(\ell)}$ on the real signal and the upper
 117 bound on the random contribution $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_j \|\mathbf{v}_{r_j}^{(\ell)}\|$. The *magnitude* of RSP-induced
 118 variation thus reduces to one number in $[0, 1]$. Early in decoding the KV cache is short and this value
 119 is non-negligible, so a single RSP draw can perturb next-token decisions.

120 The scalar w alone, however, controls only *how much* and not *where* the perturbation lands; the
 121 direction is decided by the distribution of $\mathbf{H}$. We isolate one position: with $\bar{h} \in \mathbb{R}^d$ a generic single
 122 centered RSP token (the i.i.d. marginal of any $\bar{h}_j$ in Eq. (1)), let $z_i(\bar{h})$ be the under-RSP output logit
 123 at vocab token i as a function of *that* position’s perturbation, with the remaining $L - 1$ centered
 124 tokens fixed at 0, and write the vocab-logit gap as $\Delta_{ij}(\bar{h}) := z_i(\bar{h}) - z_j(\bar{h})$ ($\bar{h} = 0$ is the centered
 125 RSP at its mean, not the no-RSP state; Appendix C separates $z^{\text{Base}}, z^{\text{RSP}}$). The transformer’s logit
 126 map is non-linear in $\bar{h}$, but a first-order Taylor expansion around $\bar{h} = 0$ gives the local surrogate
 127 $\Delta_{ij}(\bar{h}) \approx \Delta_{ij}(0) + b_{ij}^\top \bar{h}$, with $b_{ij} := \nabla_{\bar{h}}(z_i - z_j)|_{\bar{h}=0} \in \mathbb{R}^d$ the gradient direction along which
 128 tokens i, j swap rank (not unit-norm in general). Since RSP is training-free, we cannot know in
 129 advance which b_{ij} opens a useful branch on a given problem, and this missing prior constrains the
 130 distribution design. A deterministic injected sequence pins its contribution to a single direction and
 131 leaves the orthogonal complement unperturbed. Uniform sampling from the vocabulary is random
 132 but inherits the anisotropic geometry of the token-embedding table, so some directions remain
 133 systematically under-perturbed. With no prior to lean on, the prudent design is to maximize the
 134 worst-case directional variance under a fixed variance budget; that solution is uniquely isotropic.

135 **Proposition 1** (Maximin directional coverage). *Let $\mathcal{D}_\rho$ denote the family of zero-mean distribu-*
 136 *tions on $\mathbb{R}^d$ with covariance Σ_D satisfying the trace budget $\text{tr}(\Sigma_D) \leq \rho^2$. For each $D \in \mathcal{D}_\rho$,*
 137 *$\min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h) = \lambda_{\min}(\Sigma_D)$. The maximin worst-case directional variance equals ρ^2/d ,*

$$\sup_{D \in \mathcal{D}_\rho} \min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h) = \rho^2/d,$$

138 *and this supremum is attained if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

139 This is a *design criterion* for the prompt law, not a guarantee of correctness; correctness enters
 140 through the task-side $p_{\min}(x)$ assumption of §3.3 and independent resampling.²

141 Gaussian RSP satisfies the equality with $\Sigma = \sigma_E^2 \mathbf{I}_d$ (so $\rho^2 = d\sigma_E^2$) and adds *full support* on $\mathbb{R}^d$
 142 (Appendix C). Isotropy ensures no direction is systematically excluded; full support gives positive
 143 probability to the open branching-event region (§3.1) whenever it is non-empty (Appendix D.3.3).
 144 Monotone temperature preserves ranking and cannot reach this region, so RSP’s *rank-changing*
 145 branching follows from the two properties together, splitting the hidden state into multiple branching
 146 states early in reasoning.

147 3.3 Annealing: KV cache growth dilutes RSP attention

148 The early branching that RSP introduces stabilizes naturally as decoding proceeds: the upper bound
 149 on the same scalar $w_{r,i}^{(\ell)}$ defined in §3.2 narrows along the sequence axis with autoregressive KV cache
 150 growth. Each step appends one real-token term while the number of RSP terms is fixed; freezing the
 151 attention-logit gap between real and RSP tokens, this asymmetry yields the following exact bound.

²The maximin is stated in input space; isotropy is not claimed to be preserved through the transformer. Any local logit direction pulls back via the model Jacobian to an input-space direction b , and isotropy guarantees only that no such pulled-back direction has zero projected variance.

Theorem 1 (Attention-mass decay under KV cache growth). Consider a single attention head with per-head key dimension d_{head} , where query $\mathbf{q}_i$ at position i attends $n \geq 1$ real keys $\mathbf{k}_j$ and $L \geq 1$ RSP keys $\mathbf{k}_{r,j'}$ under finite logits. Define the pre-scale attention-logit gap $\Delta_i := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [L]} \mathbf{q}_i^\top \mathbf{k}_{r,j'}$ (distinct from the vocab-logit gap Δ_{ij} of §3.2). In scaled dot-product attention,

$$w_{r,i} \leq \frac{L}{L + n \exp(\Delta_i / \sqrt{d_{\text{head}}})},$$

and the right-hand side is strictly decreasing in n and tends to 0 at fixed L and Δ_i .

Proof. Let $s_j := \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_{\text{head}}}$, $s_{r,j'} := \mathbf{q}_i^\top \mathbf{k}_{r,j'} / \sqrt{d_{\text{head}}}$, $s_{r,\max} := \max_{j'} s_{r,j'}$, and $R := \sum_{j'=1}^L \exp(s_{r,j'})$. The gap definition gives $s_j \geq \Delta_i / \sqrt{d_{\text{head}}} + s_{r,\max}$ for every real j , and $\exp(s_{r,\max}) \geq R/L$ holds because the max dominates the average; combining the two and summing over the n real keys yields $\sum_{j=1}^n \exp(s_j) \geq (n/L) \exp(\Delta_i / \sqrt{d_{\text{head}}}) R$. Substituting into $w_{r,i} = R / (\sum_j \exp(s_j) + R)$ gives

$$w_{r,i} \leq \frac{R}{(n/L) \exp(\Delta_i / \sqrt{d_{\text{head}}}) R + R} = \frac{L}{L + n \exp(\Delta_i / \sqrt{d_{\text{head}}})}.$$

The derivative of the right-hand side in n is $-L \exp(\Delta_i / \sqrt{d_{\text{head}}}) / (L + n \exp(\Delta_i / \sqrt{d_{\text{head}}}))^2 < 0$, so the bound is strictly decreasing and tends to 0 as $n \rightarrow \infty$. $\square$

Applied at each layer ℓ to $w_{r,i}^{(\ell)}$ of §3.2 with gap $\Delta_i^{(\ell)}$: L caps maximum influence while n grows each step, narrowing the upper bound at fixed L . The theorem guarantees only sequence-axis attenuation at a fixed (layer, query, attention-logit gap); it does not assert layer-axis monotone decay. The gap $\Delta_i^{(\ell)}$ does evolve during decoding but tends to sharpen at deeper layers and longer sequences as real-token correlations accumulate, with Figure 2(a)’s layer-axis attenuation as indirect evidence. The accurate reading is “the attainable upper bound at a fixed gap decreases monotonically,” not “realized mass decreases per step.” Since Eq. (2) ties the random contribution to the same scalar, branching-event reachability (§3.1) inherits the envelope: as the cache grows, new branching events become rarer and the response commits to the branch already opened, yielding an implicit explore-then-exploit. Figure 2 confirms how this envelope manifests: the sequence-axis decrease matches Theorem 1’s prediction directly, while the layer-axis attenuation is observed empirically and lies outside the theorem’s scope.

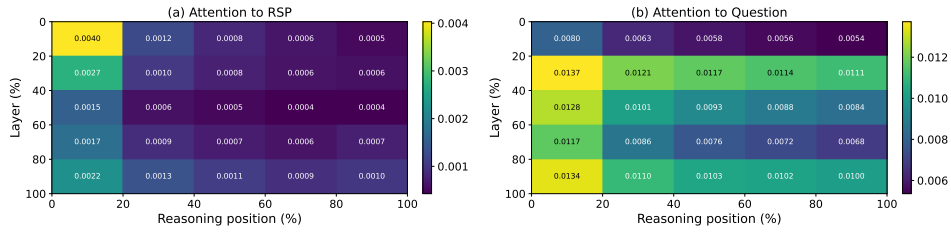


Figure 2: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). (a) RSP-token attention decreases along the reasoning-position axis, consistent with the KV cache-growth bound of Theorem 1. The additional decrease along the layer axis is a separate empirical phenomenon suggesting that deeper layers sharpen the real-vs-random attention-logit gap; it does not follow from Theorem 1 alone. (b) Question-token attention remains comparatively uniform. The reported quantity is the per-token mass of Appendix E, equal to $w_{r,i}^{(\ell)} / L$ averaged over heads and samples.

Lifting this single-shot branching to task accuracy via independent re-sampling across rollouts (§3.1) requires two assumptions: (M1) one execution reaches a correct trajectory with marginal probability $p_{\min}(x) > 0$, and (M2) the N executions are mutually independent. Under both, a standard ensemble argument yields $\Pr(\text{Pass}@N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N$ (Appendix D.3.4). §3.2 showed that isotropy and full support together make single-shot branching reachable; *independent resampling* now lifts those single-shot branches into an N -fold cumulative gain. Sharing one RSP across samples breaks (M2) and removes the accumulation, so independence is the key, and §4 verifies this signature empirically.

Table 1: Accuracy (%) across model configurations and benchmarks. Values in parentheses indicate the difference from the baseline. Full per-position (Prefix, Suffix) breakdown is in Appendix A.

Model	MATH-500			GSM8K			AIME24		
	Baseline	RSP(prefix)	RSP(suffix)	Baseline	RSP(prefix)	RSP(suffix)	Baseline	RSP(prefix)	RSP(suffix)
<i>Instruct Models</i>									
LLaMA-3.1-8B-Inst	52.20	45.00 (−7.2)	49.40 (−2.8)	85.75	76.35 (−9.4)	86.73 (+1.0)	6.67	3.33 (−3.3)	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	81.40 (−1.8)	83.40 (+0.2)	95.45	94.92 (−0.5)	95.68 (+0.2)	13.33	13.33 (+0.0)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	74.80 (+1.8)	74.20 (+1.2)	85.29	85.29 (+0.0)	84.69 (−0.6)	10.00	13.33 (+3.3)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>									
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	55.60 (−16.6)	84.15	84.31 (+0.2)	54.13 (−30.0)	6.67	16.67 (+10.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	54.60 (−6.8)	77.86	74.30 (−3.6)	58.61 (−19.3)	16.67	10.00 (−6.7)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>									
Qwen2.5-Math-7B	52.20	59.20 (+7.0)	70.40 (+18.2)	58.30	56.41 (−1.9)	75.97 (+17.7)	23.33	3.33 (−20.0)	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	51.00 (+16.6)	37.45	67.02 (+29.6)	50.64 (+13.2)	13.33	20.00 (+6.7)	13.33 (+0.0)

4 Empirical Validation

We examine how the *early hidden-state branching followed by KV cache-growth stabilization* pattern of Figure 1 surfaces in LLM generation through three scenes: whether RSPs preserve baseline accuracy and even recover it in some settings (§4.2); whether the early diversification and later stabilization suggested by the attention decomposition and the KV cache-growth envelope are observed in practice (§4.3); and whether Pass@ N depends on *independence* of the RSP draw across samples (§4.4).

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates, for seven configurations in total. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity, and the answer-extraction pipeline uses fallbacks to score only the final answer. RSPs follow the definition of §3.1 and are concatenated at one of three positions (prefix, infix, suffix) with a freshly drawn $\mathbf{H}$ for each example. The position and length L reported in Tables 1 and 2 are the best of $L \in \{10, 15, 20\}$ on the evaluation set, so these accuracy comparisons are *exploratory* and held-out hyperparameter selection is left for future work; the mechanism analyses in §4.3–§4.4 use a fixed setting ($L=10$, suffix), which insulates them from this selection effect. Full experimental details and per-position results are in Appendix A.

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects reasoning performance. Table 1 reports accuracy at the Prefix and Suffix positions (the full per-position breakdown is in Appendix A).

On instruct models RSP stays within a few percentage points of the baseline. The most striking pattern is format-mismatch recovery: when a ChatML template is applied to a base model not trained for that format, Suffix recovers +18.2/ +17.7 pp on Qwen2.5-Math-7B (MATH-500/GSM8K) and Prefix recovers up to +29.0/ +29.6 pp on Qwen2.5-Math-1.5B; if simple noise were responsible, accuracy should have dropped further. In the base raw-text setting the picture inverts: Suffix incurs sharp losses (−16.6/ −30.0 pp on Qwen2.5-Math-7B) while Prefix barely changes accuracy, consistent with a hypothesis that base models treat random signals adjacent to the EOS as a termination/derailment trigger that Prefix avoids.

RSP’s *direct accuracy gains* thus concentrate in misaligned-input recovery and stay within $\pm$ a few percentage points on well-aligned instruct models. The effects this paper centers on are the token-, trajectory-, and outcome-level diversity documented in §4.4, with accuracy recovery as a secondary outcome (extending to additional seeds is left for future work).

We further compare RSP with TTTSV, SoftCoT, and LTPO, three prior soft prompt methods that optimize embeddings under different objectives. Table 2 reports a unified evaluation under the same fallback-based answer-extraction pipeline (Appendix A.2). RSP does not consistently dominate

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. **Baseline** from Table 1; **RSP** is the best over positions and $L \in \{10, 15, 20\}$ (breakdown in Appendix A). **Bold**: best; underline: second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	<u>83.80</u>	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	95.30	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

the trained methods; TTSV, for instance, beats RSP on AIME24 with Qwen2.5-Math-7B-Instruct (23.30% vs 16.67%). The point we want to make is the converse: even without training, RSP reaches a comparable range, and that fact is itself this paper’s main evidence that part of the soft-prompt effect comes from the *structure of injection* rather than the learned content.

4.3 How Does RSP Affect the Output Distribution?

We next examine how RSP injection reshapes the output distribution during generation. The setting is Qwen2.5-Math-1.5B-Instruct on MATH-500, and we measure per-token entropy, top-1 probability, and varentropy across generation steps. Figure 3 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV); full curves are in Appendix A.4.

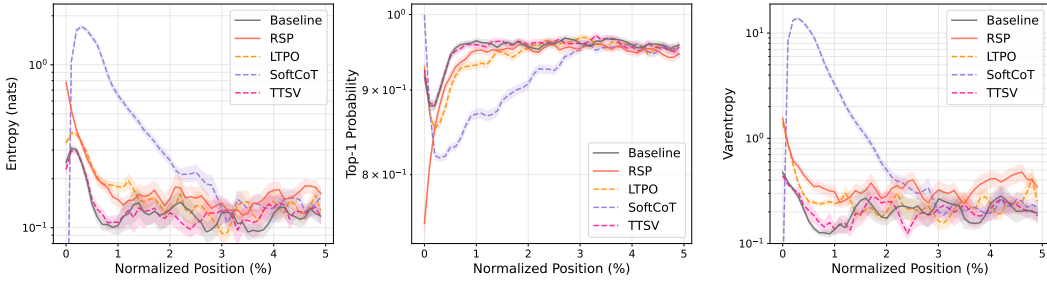


Figure 3: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

Latent prompt methods including RSP share the same signature: *a rise in early-generation entropy followed by convergence to the baseline later in generation*. A natural mechanistic explanation is the out-of-vocabulary (OOV) effect. When continuous embeddings outside the vocabulary enter the input, the model has to attend to a never-before-seen key position on top of its familiar token distribution, flattening the early next-token distribution. All three RSP injection positions show early-stage entropy $\uparrow$, top-1 probability $\downarrow$, and varentropy $\uparrow$, indicating multimodal distributions [Ahmed et al., 2026]. The change is confined to the early stage and all three metrics return to baseline in the middle and final portions, consistent with the *early influence with automatic KV cache-growth decay* pattern predicted in §3.3.

Existing methods trained under different objectives (LTPO, SoftCoT, TTSV) share this signature. Even TTSV, whose reward explicitly *lowers* mean reasoning-trajectory entropy, shows early-stage entropy comparable to or higher than the baseline. The entropy value itself is not a quality score; higher entropy is not always better. What the signature reveals is a *shared mechanism* in which OOV embedding injection produces the same fingerprint independently of the optimization objective, supporting the hypothesis that the key driver of latent reasoning is the *act of injection*, not the learned content.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature.

We quantify how much RSP shifts the first-token distribution, the critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at $\tau=1.0$ as reference, we compare baselines at $\tau=2.0, 3.0$ (16 samples per problem) against RSP at $\tau=1.0$ (averaged over 16 seeds) on four metrics: Spearman rank correlation ρ , probability mass outside the baseline’s top-10, Jensen–Shannon divergence, and Pass@1 (definitions in Appendix B). Table 3 shows the contrast: RSP’s distributional shift falls between $\tau=2.0$ and $\tau=3.0$ in magnitude, but the mechanism differs. Temperature is a monotone transform that preserves ranking ($\rho = 1$ at any τ), while RSP partially preserves but reranks ($\rho = 0.709$). The Pass@1 row exposes the cost: matching RSP’s magnitude with temperature toward $\tau=3.0$ collapses accuracy to 1.42%, whereas RSP preserves 73.30%. RSP produces a fundamentally different perturbation than temperature scaling.

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing baselines at higher temperatures ($\tau=2.0, 3.0$) with RSP at $\tau=1.0$. RSP metrics are averaged over 16 random seeds; Acc reports mean $\pm$ std across 16 samples per problem.

Metric	$\tau=2.0$	$\tau=3.0$	RSP
Spearman ρ	1.000	1.000	0.709
Mass outside top-10	5.18%	29.11%	21.86%
JS divergence	0.060	0.218	0.131
Acc (%)	20.77 $\pm$ 1.47	1.42 $\pm$ 0.35	73.30 $\pm$ 1.07

Trajectory-level: semantic diversity.

Does first-token reranking translate into semantically diverse trajectories? We sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at $\tau = 1.0$. Each trajectory is encoded by BGE-M3 [Chen et al., 2024] into a dense semantic vector, and we compute three metrics: *pairwise cosine distance* (overall semantic spread), *inter-cluster distance* between DBSCAN [Ester et al., 1996] centroids (separation between groups), and *intra-cluster distance* (coherence within groups). Table 4 reveals three concurrent patterns: pairwise distance rises by 1.4 $\times$, inter-cluster distance jumps by 5.4 $\times$, and intra-cluster distance is essentially unchanged. The trajectory set becomes more diverse and splits into more separated groups while preserving within-group coherence. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	0.0879
Inter-cluster dist.	0.0183	0.0982
Intra-cluster dist.	0.0526	0.0528

Outcome-level: Pass@ N scaling.

Does this token- and trajectory-level diversity translate into task-level outcome diversity, measured by Pass@ N [Chen et al., 2021] (the probability that at least one of N sampled solutions is correct)? We compare four conditions on MATH-500 and AIME24 with $N = 16$ samples per problem: (i) baseline at $\tau = 1.0$; (ii) a single RSP shared across all samples at $\tau = 1.0$; (iii) a fresh RSP per sample with greedy decoding; and (iv) a fresh RSP per sample at $\tau = 1.0$.

Condition (iv) achieves the highest Pass@ N on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $N = 16$, while (ii) shows no improvement over the baseline. The diversity gain depends on varying the embeddings across samples rather than fixing a single perturbation. Pass@1 stays comparable to the baseline on MATH-500, and on the harder AIME24 (iv) even surpasses it, suggesting that independent RSPs combined with temperature sampling expand trajectory diversity without sacrificing single-sample accuracy.

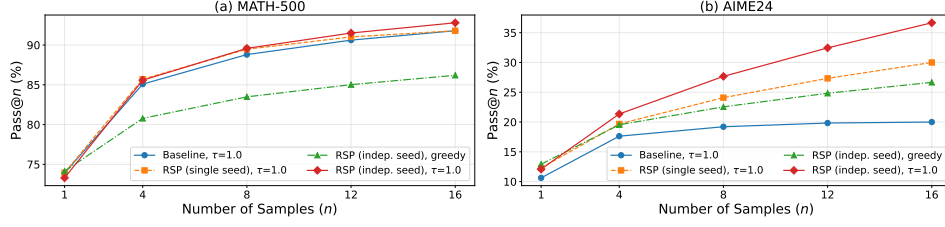


Figure 4: Pass@ N scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $N = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

5 Application: DAPO Training with RSP

Sections 3–4 established what RSP does (attention-mediated branch opening) and how it behaves at inference (Pass@ N gains, rank-changing diversity unattainable by temperature). Can we use the same effect at training time? Rollout diversity drives learning in sampling-based RL, and DAPO [Yu et al., 2025] is a recent GRPO [Shao et al., 2024] variant explicitly designed to preserve diversity at the loss level. We use it as a stringent testbed: does input-side branch opening compose with loss-side preservation to yield further gains? Details and per-step results are in Appendix F.

DAPO preserves rollout diversity at the loss level by reshaping how rollouts are weighted and clipped during long chain-of-thought RL; RSP induces diversity *before* the loss is computed by perturbing the rollout distribution at the input level (§4.4). Writing each rollout’s token sequence as $y_i = (y_{i,1}, \dots, y_{i,|y_i|})$ (using y to avoid collision with the head output $\mathbf{o}_i^{(\ell)}$ of §3.2) and pairing each rollout with an independently sampled RSP $\mathbf{H}_i$ (reused at update time to keep the update on-policy), importance ratio $r_{i,t}(\theta) = \pi_\theta(y_{i,t} | q, \mathbf{H}_i, y_{i,<t}) / \pi_{\theta_{\text{old}}}(y_{i,t} | q, \mathbf{H}_i, y_{i,<t})$, group-relative advantage $\hat{A}_{i,t}$, and asymmetric clip range $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$, the DAPO + RSP objective is

$$\mathcal{J}_{\text{DAPO+RSP}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |y_i|} \sum_{i,t} \min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t}) \right]. \quad (3)$$

Table 5: Per-benchmark accuracy (%) at each method’s peak step on Qwen2.5-Math-7B. **Bold** denotes the better of the two RL methods on each benchmark. Avg is the unweighted five-benchmark mean.

Method	GSM8K	MATH-500	College	Minerva	AIME 24	Avg
Base	57.2	52.6	18.3	13.6	8.6	30.06
DAPO	86.3	78.2	41.4	37.5	22.6	53.20
DAPO + RSP	87.7	78.6	42.2	39.7	23.6	54.36

We train Qwen2.5-Math-7B on a MATH Level-3–5 subset (~8.5K prompts) with SimpleRL-Zoo [Zeng et al., 2025] under Eq. (3): each step uses $G = 8$ rollouts per prompt at $\tau=1.0$, batch 1,024, 4 PPO mini-batches of 256, for 100 steps. DAPO is compared against DAPO + RSP, where suffix RSP ($L=20$) is applied during rollouts only and evaluation uses no RSP. We evaluate on five math benchmarks (GSM8K [Cobbe et al., 2021], MATH-500 [Lightman et al., 2024], College Math, Minerva Math [Lewkowycz et al., 2022], AIME 2024) and report the unweighted mean; AIME 2024 uses Avg@32 at $\tau=1.0$, the others greedy.

Figure 5 and Table 5 show two patterns: a late-stage advantage and a delayed early reward growth. DAPO + RSP reaches 54.36% at step 90 (vs DAPO’s 53.20% peak, +1.16 pp) and widens the gap to +3.10 pp at step 100 with DAPO’s post-peak decline delayed; earlier the curves cross around step 20. The patterns follow from the methods acting at separate stages: DAPO preserves diversity at the loss level over already-generated

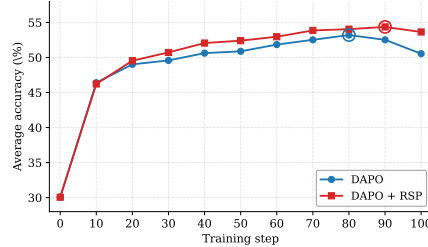


Figure 5: Five-benchmark average accuracy on Qwen2.5-Math-7B. DAPO + RSP reaches a higher peak (step 90) and stays stable through step 100.

rollouts while RSP perturbs the hidden states that produce them (§3.2), exposing the policy to broader learning signals. This is the exploration–exploitation dynamics of RL [Sutton and Barto, 2018] induced from the input. These results use a single training seed and should be read as a feasibility demonstration; multi-seed evaluation is future work.

6 Conclusion

We studied Random Soft Prompts (RSPs), training-free isotropic Gaussian vectors fitted to pretrained embedding statistics. The RSP contribution per attention layer reduces to a single scalar $w_{r,i}^{(\ell)}$ whose fixed-gap upper bound shrinks under KV cache growth, and isotropic resampling reaches top- K entry regions that rank-preserving temperature cannot (§3). Latent-prompt methods share the same entropy signature, the gain vanishes when one RSP is shared across samples (§4), and DAPO + RSP yields a modest late-training advantage (§5). This signature emerges from purely random input-side perturbation, suggesting that part of what trained soft prompts have been credited with is a *structural by-product of injection*, and that RSP composes orthogonally with sampling-based pipelines as a near-zero-cost diversity expander. Open questions remain on generalization beyond RoPE decoders and math reasoning, deep-layer sharpening outside Theorem 1’s envelope, multi-seed DAPO variance, and held-out hyperparameter selection (§4.1).

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A Experimental Setup and Full Accuracy Breakdown

Table 6 provides the full per-position accuracy breakdown (Prefix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	6.67
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	<u>13.33</u> (+0.0)
	Suffix	83.40 (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	<u>85.29</u> (+0.0)	<u>13.33</u> (+3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	<u>84.15</u>	6.67
	Prefix	71.60 (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	74.30 (−3.6)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	<u>3.33</u> (−20.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. **Blue text** indicates RSP embeddings, and **purple text** indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

545 **ChatML (Qwen Instruct / Base + ChatML).**

```
[Random Embeddings ( $L$  tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

547 **LLaMA Chat Template.**

```
[Random Embeddings ( $L$  tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

549 **Raw Text (Qwen Base).**

```
[Random Embeddings ( $L$  tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
```

551 **A.1.2 Infix**

552 A description text and special tokens are inserted into the prompt. The special token positions are
553 then replaced with RSP embeddings at the embedding level.

554 **ChatML (Qwen Instruct / Base + ChatML).**

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}

There are { $L$ } special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the { $L$ } special tokens: <special_token_1>...<special_token_ $L$ >
<|im_end|>
<|im_start|>assistant
```

556 **LLaMA Chat Template.**

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}

There are { $L$ } special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the { $L$ } special tokens:
<reserved_special_token_0>...
<reserved_special_token_ $L$ >
```



```
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}

There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the {L} special tokens: <special_token_1>...<special_token_L>

Please reason step by step, and put your final answer within \boxed{}
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.
For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. \boxed{...}: last non-empty box is used; empty \boxed{} are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 6 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 9 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 9: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle/last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 3 (main text) and Figure 6.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

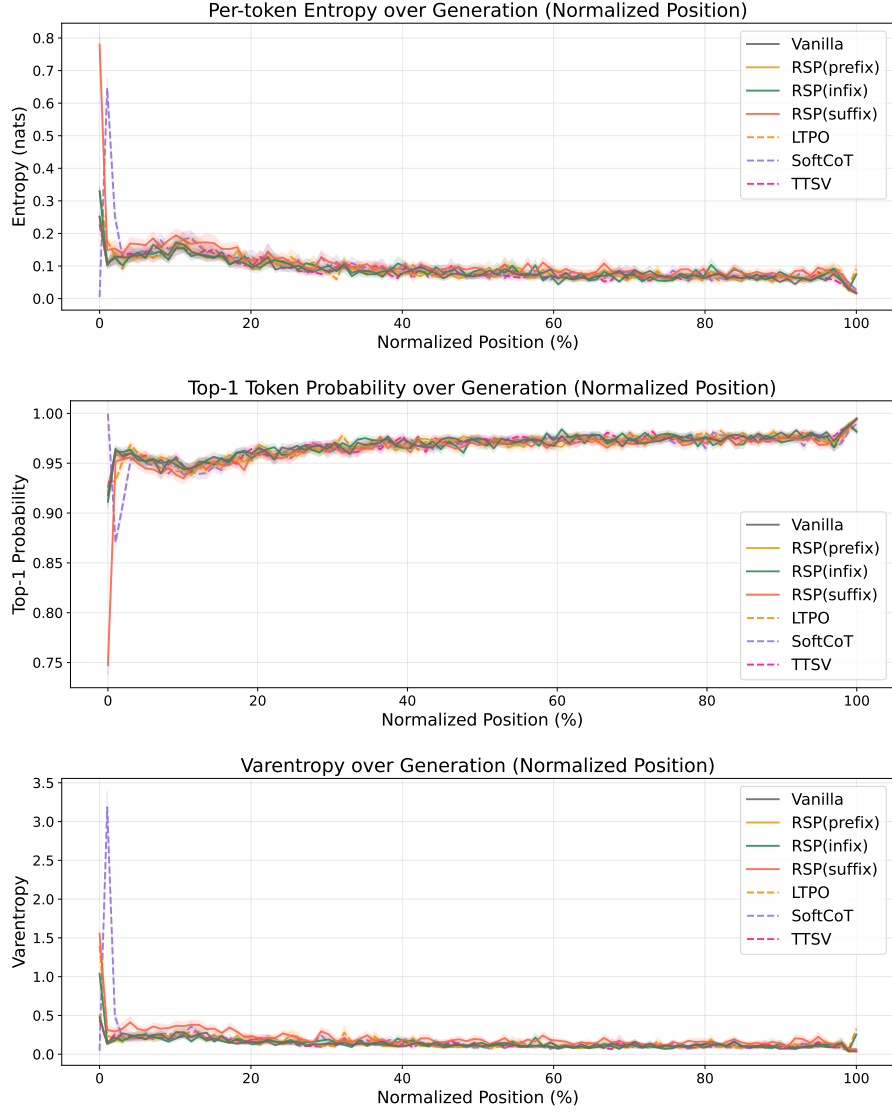


Figure 6: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

596 B Token-Level Distribution Metrics

597 This appendix provides formal definitions for the metrics used in Section 4.4. Numerical results
 598 are reported in Table 3 of the main text. We extract first-token logits via a single forward pass (no
 599 autoregressive generation) and store the top-100 token ids and logit values per problem. Temperature
 600 conditions reuse the baseline logits scaled by $1/\tau$, requiring no separate inference. RSP draws
 601 16 independent seeds per problem (MATH-500), and reported metrics are the mean across the 16
 602 seeds. Pass@1 accuracy is computed over 16 samples per problem, with the baseline at $\tau=1.0$
 603 reaching $73.92 \pm 0.78\%$. Let P_{base} denote the baseline next-token distribution at $\tau=1.0$ and P_{target} the
 604 candidate distribution being compared (e.g., baselines at $\tau=2.0, 3.0$ or RSP at $\tau=1.0$). All metrics
 605 are computed analytically from saved logits at the first generation step.

606 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 607 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (4)$$

608 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 609 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 610 indicates rank reorganization beyond what temperature scaling can produce.

611 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 612 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (5)$$

613 This directly measures support expansion: how much probability the candidate assigns to tokens that
 614 are not in the baseline’s preferred set. Reported values use $K = 10$.

615 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (6)$$

616 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 617 sentinel logit (-10^9) before softmax, which is negligible for our setting.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 3.2 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let z_i^{Base} denote the no-RSP output logit at vocab token i and $z_i^{\text{RSP}}(\bar{h})$ the corresponding logit when the centered RSP perturbation takes value $\bar{h}$. The (true) vocab-logit gap under RSP is $\Delta_{ij}(\bar{h}) := z_i^{\text{RSP}}(\bar{h}) - z_j^{\text{RSP}}(\bar{h})$, distinct from the no-RSP baseline gap $\Delta_{ij}^{\text{Base}} := z_i^{\text{Base}} - z_j^{\text{Base}}$. Note that $\bar{h} = 0$ corresponds to the centered RSP at its entrywise mean rather than to the no-RSP forward pass, so in general $\Delta_{ij}(0) \neq \Delta_{ij}^{\text{Base}}$. The transformer’s logit map is non-linear in $\bar{h}$, but a first-order Taylor expansion around $\bar{h} = 0$ yields the local surrogate

$$\Delta_{ij}^{\text{lin}}(\bar{h}) := \Delta_{ij} + b_{ij}^\top \bar{h},$$

with $\Delta_{ij} := \Delta_{ij}(0)$ and $b_{ij} := \nabla_{\bar{h}}(z_i^{\text{RSP}} - z_j^{\text{RSP}})|_{\bar{h}=0}$. The proposition below is stated for the surrogate Δ_{ij}^{lin} . For the true non-linear gap $\Delta_{ij}(\bar{h})$, the first-order term has zero mean under centered perturbations, but the Taylor remainder $r_{ij}(\bar{h}) = O(\|\bar{h}\|^2)$ may contribute a second-order mean shift; we make no claim about its sign or magnitude.

Proposition 2 (No systematic first-order drift in the surrogate). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}[\Delta_{ij}^{\text{lin}}(\bar{h})] = \Delta_{ij}$$

for every token pair (i, j) . If a deterministic centered prompt $\bar{h}_0$ is reused across runs, then

$$\Delta_{ij}^{\text{lin}}(\bar{h}_0) - \Delta_{ij} = b_{ij}^\top \bar{h}_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}[\Delta_{ij}^{\text{lin}}(\bar{h})] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. $\square$

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Proposition 1

For a unit vector b , the first-order perturbation energy available along b is $\text{Var}_{h \sim D}(b^\top h) = b^\top \Sigma_D b$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D .

Proof. For any symmetric covariance matrix Σ_D ,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d} \sum_{m=1}^d \lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. $\square$

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

649 **C.3 Open-Set Reachability of Isotropic Gaussian Prompts**

650 **Proposition 3** (Open-set reachability). *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ with $\sigma > 0$. Then every nonempty open*
 651 *set $U \subseteq \mathbb{R}^d$ satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

652 *Proof.* The Gaussian density

$$(2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

653 is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue
 654 measure, and integrating a strictly positive density over that ball gives positive probability. $\square$

655 This is the only support property used in the informal branching argument of §3.2 and in Ap-
 656 pendix D.3.3.

657 D Proofs for the Transformer-Level Mechanism

658 This appendix follows the same order as the main theory section: exploration, annealing, and
659 performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

660 D.1 Exploration: attention decomposition and perturbation size

661 **Proposition 4** (Attention decomposition). *For one attention head at query position i in layer ℓ , let*
662 *$\alpha_{ij}^{(\ell)}$ be the attention weights over $n \geq 1$ unmasked real tokens and $L \geq 1$ random tokens, with finite*
663 *attention logits. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$$

664 and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

665 Then

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)},$$

666 where

$$\tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \quad \text{and} \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

667 *Proof.* The head output is

$$\mathbf{o}_i^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

668 By the softmax positivity and the existence of at least one unmasked real token, $\sum_{j=1}^n \alpha_{ij}^{(\ell)} =$
669 $1 - w_{r,i}^{(\ell)} > 0$, so

$$\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)}.$$

670 Substituting this identity gives the decomposition. □

671 **Corollary 1** (Magnitude scales with random-token attention). *For every realization of the random*
672 *values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

673 *Proof.* By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \leq \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

674 □

675 Corollary 1 is the only quantitative fact needed in the main text. No independence assumption
676 between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution
677 must be small as well.

678 D.2 Corollaries of Theorem 1 on KV cache growth

679 The fixed-gap decay bound yields the corollaries below.

680 **Corollary 2** (Sequence-wise annealing of the fixed-gap envelope). *For fixed L and $\Delta_i^{(\ell)}$, the upper*
 681 *bound*

$$f(n) = \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d_{\text{head}}})}$$

682 *is strictly decreasing in n and tends to 0 as $n \rightarrow \infty$. KV cache growth alone therefore narrows the*
 683 *largest RSP attention mass compatible with that gap.*

684 During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve.
 685 The accurate reading is not “the realized mass decreases at every step” but “the envelope compatible
 686 with a fixed gap shrinks as the KV cache grows.”

687 *Proof.* Differentiate $f(n)$ with respect to n :

$$f'(n) = -\frac{L \exp(\Delta_i^{(\ell)} / \sqrt{d_{\text{head}}})}{\left(L + n \exp(\Delta_i^{(\ell)} / \sqrt{d_{\text{head}}})\right)^2} < 0.$$

688 Thus $f(n)$ is strictly decreasing, and its denominator diverges linearly in n while its numerator is
 689 fixed, so $f(n) \rightarrow 0$. $\square$

690 **Corollary 3** (Vanishing random contribution under annealing). *If $n \rightarrow \infty$ while L and $\Delta_i^{(\ell)}$ remain*
 691 *fixed, $w_{r,i}^{(\ell)} \rightarrow 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \rightarrow 0.$$

692 *Proof.* $w_{r,i}^{(\ell)} \rightarrow 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite
 693 for any realization of finitely many sampled values. $\square$

694 D.3 Performance gain: from local admission to Pass@N

695 From here we work at the vocab-logit and task levels, dropping the per-layer index ℓ . The main text
 696 uses a first-order surrogate to state two claims: the §3.2 claim that a baseline-excluded vocab token
 697 can enter the local top- K set, and the §3.3 closing that independent reruns turn such local openings
 698 into Pass@ N gains. The proofs below use only one support condition on the centered prompt law D :

$$(S1) \quad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

699 Proposition 3 shows that the isotropic Gaussian law of §3.2 satisfies (S1).

700 D.3.1 Autoregressive Formulation

701 In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

702 where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by
 703 RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single
 704 global distribution.

705 **D.3.2 Local Linearization of Logits under RSP**

706 Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition
 707 (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length- L prompt
 708 $\mathbf{H} = (h_1, \dots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation
 709 can shift the logits. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in
 710 a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_{\bar{h}} z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

711 where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_{\bar{h}} z_i(0)$ and $z_i := z_i(0)$, we obtain the local affine
 712 surrogate

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

713 This approximation is used as a first-order surrogate for branch opening, not as an exact global model
 714 of the transformer logits.

715 **D.3.3 Top- K entry region (Remark)**

716 This complements the informal branching argument in §3.2. Define the entry region

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

717 If $\mathcal{G}_{i,K}$ contains a nonempty open set U , then by (S1) $D(U) > 0$, and $U \subseteq \mathcal{G}_{i,K}$ gives $\Pr_{\bar{h}}(\bar{h} \in$
 718 $\mathcal{G}_{i,K}) \geq D(U) > 0$. A useful sufficient condition is the existence of some $\bar{h}_0$ at which token i strictly
 719 outranks all but at most $K - 1$ other tokens in the affine surrogate; by continuity, a neighborhood of
 720 $\bar{h}_0$ is then contained in $\mathcal{G}_{i,K}$. This is a rank-changing event unreachable by the monotone temperature
 721 transform, but it does not by itself imply (M1).

722 **D.3.4 Pass@ N ensemble bound (Remark)**

723 The bound $\Pr(\text{Pass@}N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N$ in the main text is a standard independent-
 724 Bernoulli union argument. Let A_n be the event that the n -th run reaches a correct trajectory on x ;
 725 under (M1) $\Pr(A_n) \geq p_{\min}(x)$, and under (M2)

$$\Pr\left(\bigcap_{n=1}^N A_n^c\right) \leq (1 - p_{\min}(x))^N, \quad \Pr\left(\bigcup_{n=1}^N A_n\right) \geq 1 - (1 - p_{\min}(x))^N \geq 1 - e^{-N p_{\min}(x)}.$$

726 This is a generic fact, not specific to RSP; RSP's role is restricted to supplying mechanisms that
 727 satisfy (M1) and (M2).

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 2 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,a)}$ denote the post-softmax attention weight at layer ℓ , head index a (using a to avoid collision with the RSP matrix $\mathbf{H}$), from query position i to key position j , satisfying $\sum_j \alpha_{i,j}^{(\ell,a)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let N_{head} be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q| N_{\text{head}}} \sum_{a=1}^{N_{\text{head}}} \sum_{j \in Q} \alpha_{i,j}^{(\ell,a)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R| N_{\text{head}}} \sum_{a=1}^{N_{\text{head}}} \sum_{j \in R} \alpha_{i,j}^{(\ell,a)}. \quad (7)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_{\bullet}^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_{\bullet}[\ell, i]. \quad (8)$$

Figure 2 reports the sample average of Eq. (8) over the 500 valid samples.

E.3 Excluding the `\boxed{\dots}` Span

The reasoning position range terminates strictly before the final `\boxed{\dots}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{\dots}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

768 E.5 Additional Suffix Heatmaps

769 Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-
 770 1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern
 771 does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across
 772 both models: question-token attention varies broadly across layers, whereas RSP attention shows
 773 sequence-wise decay consistent with Theorem 1 alongside an empirical layer-wise attenuation that
 774 the theorem itself does not predict.

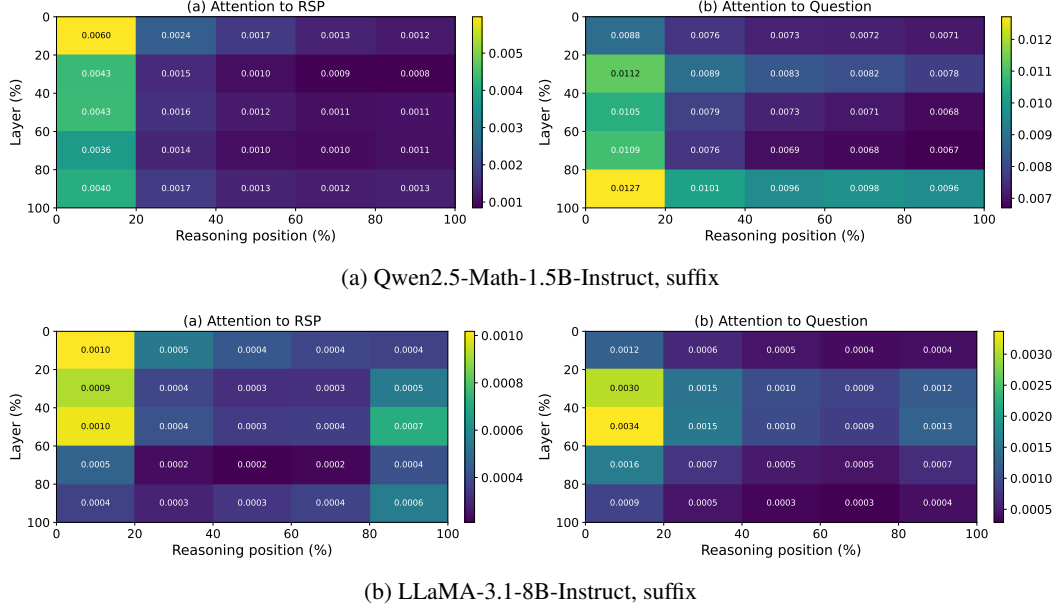


Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 2: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

775 F DAPO Implementation

776 This appendix complements Section 5 with the DAPO loss modifications relative to GRPO, the
 777 DAPO + RSP implementation, and full per-step results.

778 F.1 From GRPO to DAPO

779 The vanilla GRPO objective [Shao et al., 2024] for a group of G rollouts $\{y_i\}_{i=1}^G$ (each $y_i =$
 780 $(y_{i,1}, \dots, y_{i,|y_i|})$ a token sequence; we use y rather than o throughout this appendix to avoid collision
 781 with the head output $\mathbf{o}_i^{(\ell)}$ of §3.2) with rewards $\{R_i\}_{i=1}^G$ is

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \left(\min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t}) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right) \right], \quad (9)$$

782 with importance ratio $r_{i,t} = \pi_\theta(y_{i,t} | q, y_{i,<t}) / \pi_{\theta_{\text{old}}}(y_{i,t} | q, y_{i,<t})$, group-relative advantage $\hat{A}_{i,t} =$
 783 $(R_i - \text{mean}(\{R_i\})) / \text{std}(\{R_i\})$, clipping range ε , and KL penalty βD_{KL} against the reference π_{ref} .

784 We build on the SimpleRL-Zoo [Zeng et al., 2025] training pipeline and implement DAPO on
 785 top of VeRL [Sheng et al., 2024]. DAPO modifies Eq. (9) in four ways: (i) token-level loss
 786 aggregation $1 / \sum_i |y_i|$ instead of $1 / |y_i|$, (ii) asymmetric clipping with $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, (iii)
 787 an implementation-side negative-advantage dual-clipping safeguard $\max(L_{\text{clipped}}, c\hat{A})$ with $c = 10$
 788 (active only when $\hat{A} < 0$; omitted from Eq. (10) below for readability), and (iv) KL terms removed.
 789 The resulting DAPO objective is

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |y_i|} \sum_{i,t} \min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right]. \quad (10)$$

790 F.2 DAPO + RSP Implementation

791 For DAPO + RSP, we additionally inject a per-rollout RSP $\mathbf{H}_i \in \mathbb{R}^{L \times d}$ ($L = 20$) into each rollout.
 792 Each $\mathbf{H}_i$ is generated deterministically from a per-step rollout seed and injected via a forward hook;
 793 it is not a trainable parameter, so the learning signal is the policy adapting to arbitrary $\mathbf{H}_i$ rather than
 794 a learned prompt. Including $\mathbf{H}_i$ in the log-probabilities at both rollout and update time keeps the
 795 update on-policy with respect to the RSP-conditioned distribution (avoiding off-policy mismatch).
 796 With $G = 8$ rollouts per prompt, each rollout receives an independently drawn $\mathbf{H}_i$.

797 F.3 Data, Batch, and Hyperparameters

Table 10: DAPO / DAPO + RSP training setup on Qwen2.5-Math-7B.

Field	Value
Train data	MATH Level 3–5 subset (<code>simplelr_math_35</code> , ~8,500 prompts)
Prompt template	<code>qwen-boxed</code>
Train batch size	1024
PPO mini-batch size	256 (4 PPO updates per rollout)
Rollouts per prompt G	8
Max prompt / response length	2,028 / 2,048
Rollout sampling	temperature 1.0, top- p 1.0, top- k 50
$(\epsilon_{\text{low}}, \epsilon_{\text{high}})$	(0.2, 0.28)
Dual clip c	10.0
Loss aggregation	token-mean ($1 / \sum_i y_i $)
KL terms	disabled
Entropy coefficient	0
Optimizer	AdamW, learning rate 5×10^{-7} (constant), no warmup
Total training	~10 epochs, ≥ 80 optimization steps
Save / eval frequency	every 10 steps
RSP (DAPO + RSP only)	suffix injection, $L = 20$, freshly drawn per rollout
Hardware	4× NVIDIA B200 GPUs
Wall-clock training time	~12 hours per run

798 F.4 Evaluation Protocol

799 Each saved checkpoint is converted to HuggingFace format and evaluated using the SimpleRL-
800 Zoo `eval_math_nodes.sh` pipeline, which selects sampling parameters per benchmark to reduce
801 variance on the smaller competition sets:

Table 11: Per-benchmark evaluation protocol used by SimpleRL-Zoo `sh/eval.sh`.

Benchmark	# Problems	Temperature	n_{sampling}	Metric
AIME 2024	30	1.0	32	Avg@32
AMC 2023	40	1.0	32	Avg@32
MATH-500	500	0.0	1	accuracy (mean@1)
GSM8K	1,319	0.0	1	accuracy
OlympiadBench	675	0.0	1	accuracy
Minerva Math	272	0.0	1	accuracy
GaoKao 2023-EN	385	0.0	1	accuracy

802 The maximum number of generated tokens is 16,000 across all benchmarks. The reported average is
803 the unweighted mean over the five benchmarks (GSM8K, MATH-500, College Math, Minerva Math,
804 AIME 2024).

Table 12: Five-benchmark average accuracy (%) at every checkpoint, computed over GSM8K, MATH-500, College Math, Minerva Math, and AIME 2024. **Bold** marks each method’s peak.

Step	DAPO	DAPO + RSP
0	30.06	30.06
10	46.40	46.22
20	49.02	49.54
30	49.58	50.72
40	50.62	52.06
50	50.88	52.40
60	51.84	52.96
70	52.52	53.86
80	53.20	54.04
90	52.52	54.36
100	50.54	53.64

806 G Broader Impacts

807 This work provides empirical and theoretical analysis of how random embedding injection affects
808 the reasoning behavior of large language models. The contributions are primarily foundational,
809 with potential downstream implications for reinforcement learning-based post-training of reasoning
810 models such as GRPO.

811 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
812 based LLM post-training by providing richer learning signals within group-relative advantage estima-
813 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
814 fine-tuning, making such methods more accessible to research groups with limited resources.

815 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
816 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
817 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
818 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
819 standalone diversity source.

820 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
821 model or dataset, the direct societal impact is limited. The broader implications described above are
822 contingent on future work integrating RSP into applied training pipelines.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and §1 state four scoped claims, each tied to a section: (i) RSP’s contribution at each attention layer is captured by a single scalar with a fixed-gap upper-envelope cache-growth bound (Theorem 1, §3.3); (ii) under a local affine surrogate, isotropic re-sampling reaches top- K entry regions unattainable by rank-preserving temperature sampling (§3.2); (iii) RSP raises early-token entropy and stabilizes later, with Pass@ N gains under independent re-sampling (§4); and (iv) the diversity transfers to RL training when combined with DAPO (§5). We do not claim a theorem that directly explains the format-mismatch recovery (Table 1); the wrong-attractor hypothesis there is explicitly informal.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Limitations are noted in several places rather than in a dedicated section: (a) the multi-path Pass@ N argument requires the task-side assumption $p_{\min}(x) > 0$, flagged in §3.2 and §6; (b) the +18–29 pp recovery on the format-mismatch rows of Table 1 is empirically observed but not formally derived from any theorem in §3, with the wrong-attractor account stated as a hypothesis to be quantified in future work (§4.2); (c) Table 1 reports the best-position accuracy across {Prefix, Infix, Suffix}, and per-position variance is large with several cases below baseline (Appendix A), so position is treated as a hyperparameter rather than something we justify principally; (d) DAPO + RSP results (§5) use a single training seed under a reduced DAPO configuration that omits Dynamic Sampling and overlong reward shaping; (e) main accuracy comparisons are single-seed without bootstrap or significance tests (item 7). Future work in §6 addresses training-time comparisons, domain expansion, and principled position selection.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[Yes\]](#)

Justification: Theorem 1 (cache-growth attenuation) carries an inline proof in §3.3. Proposition 1 is stated in §3.2 with its proof in Appendix C; the supporting top- K entry and Pass@ N ensemble arguments around it are presented as informal discussion in §3.2 with formal write-ups in the appendix. Each statement explicitly lists its assumptions: finite logits and $n, L \geq 1$ for Theorem 1; finite second-moment budget for Proposition 1; and the locally affine surrogate plus $p_{\min}(x) > 0$ task-side condition together with mutual independence of the N runs for the Pass@ N argument.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: §4.1 and Appendix A specify benchmarks (MATH-500, GSM8K, AIME 2024 in the main table; further sets used for the DAPO study), models (LLaMA-3.1-8B-Instruct and Qwen2.5-Math 1.5B/7B variants), the RSP injection grid (Prefix/Infix/Suffix; $L \in \{10, 15, 20\}$), greedy decoding, and the answer extraction pipeline. Appendix F provides the full DAPO + RSP training recipe (batch of 1,024 prompts, $G = 8$ rollouts per prompt at $\tau = 1.0$, four PPO mini-batch updates of 256 prompts each, 100 training steps, suffix RSP $L = 20$ during rollouts only).

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: All datasets and base models used in the paper are publicly available and cited (MATH-500, GSM8K, AIME 2024, AMC 2023, OlympiadBench, Minerva Math, GaoKao 2023-EN, College Math; Qwen2.5-Math 1.5B/7B and LLaMA-3.1-8B-Instruct). RSP itself is described to a level of detail sufficient for re-implementation (Eq. (1) and §4.1). However, our RSP injection harness and DAPO + RSP training pipeline are not yet released; we will release an anonymized implementation alongside the camera-ready submission.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §4.1 and Appendix A list all benchmarks, models, RSP injection settings (positions, lengths, per-row L choice in Appendix Table 7), maximum generation length per benchmark, and decoding strategy. Appendix F provides the full DAPO + RSP training settings, including optimizer, batch composition, mini-batch updates, asymmetric clipping range $(\epsilon_{\text{low}}, \epsilon_{\text{high}}) = (0.2, 0.28)$, dual clipping safeguard, and RSP injection protocol during rollouts.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Pass@ N experiments use $N = 16$ samples per problem (§4.4), Table 3 reports mean±standard deviation over 16 RSP seeds, and AIME 2024 is scored by Avg@32 to reduce variance. However, Table 1 (Table 1) reports single-run accuracies without bootstrap confidence intervals or paired tests, and the DAPO + RSP training in §5 uses a single training seed. Multi-seed evaluation, particularly on small competition sets (AIME 2024, AMC 2023) where variance is highest, is left for future work.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Inference experiments (Tables 1, 2; Figure 3) run on a single NVIDIA A6000 40GB GPU with greedy decoding (Appendix A). DAPO + RSP training (§5) runs on $4 \times$ NVIDIA B200 GPUs for approximately 12 hours per run (Appendix F, hardware row of the configuration table).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses publicly available math reasoning benchmarks and pretrained language models, involves no human subjects or personal data, and does not introduce a release that bypasses standard model-safety considerations. We have reviewed the NeurIPS Code of Ethics and identify no conflicts.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix G discusses both directions: positive impact via more sample-efficient RL post-training of reasoning LLMs, and a potential downside that broadening the effective output support also increases reachability of undesirable trajectories, which we argue should keep RSP coupled with existing safety filtering rather than replacing it.

931 **11. Safeguards**

932 Question: Does the paper describe safeguards that have been put in place for responsible
933 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
934 image generators, or scraped datasets)?

935 Answer: [N/A]

936 Justification: The paper does not release new pretrained models, generative systems, or
937 scraped datasets. RSP is a generation-time perturbation applied to existing pretrained models
938 and introduces no separate artifact requiring safeguards.

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941 the paper, properly credited and are the license and terms of use explicitly mentioned and
942 properly respected?

943 Answer: [Yes]

944 Justification: All datasets (MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021],
945 AIME 2024, AMC 2023, OlympiadBench [He et al., 2024], Minerva Math [Lewkowycz
946 et al., 2022], GaoKao 2023-EN, College Math) and pretrained models (Qwen2.5-Math
947 [Yang et al., 2024], LLaMA-3.1 [Grattafiori et al., 2024]) are publicly available and cited
948 at first use. The training pipeline builds on SimpleRL-Zoo [Zeng et al., 2025] and VeRL
949 [Sheng et al., 2024], also cited. Each asset is used under its original publicly stated terms.

950 **13. New assets**

951 Question: Are new assets introduced in the paper well documented and is the documentation
952 provided alongside the assets?

953 Answer: [N/A]

954 Justification: The paper does not release new datasets or pretrained models. The implemen-
955 tation code planned for camera-ready release (item 5) will be accompanied by configuration
956 files and a README; no derived dataset or model checkpoint is part of the submission.

957 **14. Crowdsourcing and research with human subjects**

958 Question: For crowdsourcing experiments and research with human subjects, does the paper
959 include the full text of instructions given to participants and screenshots, if applicable, as
960 well as details about compensation (if any)?

961 Answer: [N/A]

962 Justification: The paper involves no crowdsourcing and no research with human subjects.
963 All evaluations use static, publicly available math reasoning benchmarks.

964 **15. Institutional review board (IRB) approvals or equivalent for research with human
965 subjects**

966 Question: Does the paper describe potential risks incurred by study participants, whether
967 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
968 approvals (or an equivalent approval/review based on the requirements of your country or
969 institution) were obtained?

970 Answer: [N/A]

971 Justification: As noted in item 14, the paper involves no human subjects, so IRB review is
972 not applicable.

973 **16. Declaration of LLM usage**

974 Question: Does the paper describe the usage of LLMs if it is an important, original, or
975 non-standard component of the core methods in this research? Note that if the LLM is used
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978

Answer: [N/A]

979

Justification: Pretrained LLMs (Qwen2.5-Math, LLaMA-3.1) appear in the paper as the experimental subject of analysis, not as a non-standard component of the methods themselves.

980

981

The methods (RSP definition, attention decomposition, DAPO + RSP training) do not rely on auxiliary LLMs in any non-standard or originality-affecting role.

982